

NetPulse

Network Capacity Analytics & Compliance Reporting Platform

NetPulse converts raw interface-utilization exports into capacity intelligence and submission-ready reports. It ingests routine polling files, computes per-link utilization, busy-period trends, growth forecasts and anomaly detection, and generates the threshold, upgrade-policy and capacity documentation that is currently assembled manually — with a complete, auditable evidence trail behind every figure. It is delivered as a single self-contained file requiring no server, no database, no licence and no network access.

Zero

INFRASTRUCTURE &
LICENCE COST

1M+

ROWS PROCESSED PER
DATASET

15–20 hrs

MANUAL EFFORT REMOVED
PER CYCLE

100%

OFFLINE — NO DATA
EGRESS

BUSINESS OUTCOMES

- **Automated capacity reporting.** Threshold-breach, upgrade-policy and capacity reports that previously required approximately **15–20 hours** of manual spreadsheet work each reporting cycle are now produced in seconds, complete with timestamped supporting evidence.
- **Submission-grade documentation.** Professional PDF reports containing reporting parameters, summary counts, per-day exceedance tables, a full evidence log and a prepared / reviewed / approved sign-off block — suitable for internal governance and regulatory submission.
- **Prevents unnecessary capital spend.** Links that appear saturated only because of stale bandwidth records are automatically flagged for verification instead of triggering an upgrade, avoiding procurement that is not required.
- **Prevents capacity-related incidents.** Genuinely congested links are identified early, with a statistical forecast of when each will exhaust its headroom and a recommended upgrade window.
- **No vendor dependency or spend.** The entire platform is one HTML file — no server, database, licence, subscription or third-party component.
- **Security and data residency by design.** Runs fully offline with all external network access blocked at the browser level; no operational data leaves the machine, avoiding the security-review and data-transfer overhead a hosted alternative would incur.
- **Reusable across the business.** Fully white-labelled and configurable — thresholds, market hours, organisation branding and user-defined fields — so the same platform serves other environments, teams and customers without code changes.

DATA-INTEGRITY ISSUES IDENTIFIED AND RESOLVED

During validation against live data the platform surfaced several material defects in the source exports. Each is now detected automatically and surfaced to the operator, so capacity decisions rest on correct figures.

ISSUE FOUND	BUSINESS CONSEQUENCE IF UNDETECTED	RESOLUTION
Stale bandwidth records — links reporting above 100% utilization	Healthy links appear saturated; unnecessary upgrade spend and false escalations	Detected and flagged for capacity review; corrected capacity applied via override, all figures recomputed
Inconsistent device naming splitting one link across two identities	Traffic under-reported; a congested link can appear quiet and be missed entirely	Automatic and manual consolidation of naming variants
Unit misinterpretation in the vendor export headers	Throughput and utilization figures materially wrong in both directions	Unit resolution corrected and validated against the source file
Timestamp inconsistency between date formats in the same export	Readings shifted in time; market-hours analysis and breach evidence unreliable	Timestamps preserved exactly as recorded, with an explicit time-zone control

ISSUE FOUND	BUSINESS CONSEQUENCE IF UNDETECTED	RESOLUTION
Threshold counts excluding off-hours activity	Overnight congestion invisible in reports	Market and non-market activity reported separately and together in one document

Net effect: every published figure is now reconcilable line-by-line against the original source file, and the conditions that previously produced misleading capacity conclusions are detected automatically rather than relying on manual review.

PLATFORM CAPABILITIES

Data ingestion

- Multi-file Excel / CSV upload; columns matched by **meaning**, not fixed names, and fully editable
- Automatic unit normalisation across bps / Kbps / Mbps / Gbps / bytes
- Scales to 6-month, 1M+ row datasets via streaming ingestion
- Append-only history; duplicate records rejected, originals never overwritten
- Validation console with parse inspector before any data is committed

Analytics

- Per-sample utilization measured against the capacity in force at that moment
- Busy-period (p95), average and peak analysis with configurable basis
- Statistical growth forecasting with confidence range and time-to-exhaustion
- Anomaly detection against each link's own behavioural baseline, with explanations
- Risk scoring weighted by business impact; market vs non-market hour comparison

Reporting

- **Threshold Breach PDF** — six-month warning and critical counts with full timestamped evidence and sign-off block
- **Upgrade Policy workbook** — summary plus per-day and per-timestamp evidence sheets
- **Regulatory Capacity Report** — utilization against a configurable threshold with headroom, augmentation trigger and projected breach date
- **Processed-data export** — every computed value exposed for independent audit
- Executive, capacity, trend and business-impact reports

Governance & configuration

- Append-only audit log of every upload, configuration change and export
- Configurable thresholds, market hours, upgrade policy and retention basis
- **User-defined custom fields** (e.g. ISP, Purpose) flowing through import, exports and reports
- Organisation branding and named prepared / reviewed / approved signatories
- Maintenance windows excluded from analytics and reporting

ARCHITECTURE, SECURITY AND DELIVERY

Architecture

A single self-contained HTML file. All processing occurs in the browser; data is held in local browser storage. There is no server, database, installation or administrative dependency — the file is opened directly and works immediately, including on an air-gapped machine.

No server No database No licence cost No installation

Air-gap capable

Security

External network access is blocked at the browser level by an enforced content-security policy. The platform makes no outbound calls of any kind — no telemetry, no external fonts, scripts or services. Operational data never leaves the machine on which it is opened.

Quality assurance

Delivered across **54 controlled releases**, each validated by an automated regression suite exercising every view, export and interaction and asserting a zero-error run. Supported by a documented operations SOP and a full change history.

PROPOSED NEXT STEPS

- Walkthrough and demonstration session with the wider network engineering team

- Extend coverage from the core links to the remaining sites and environments
- Confirm reporting thresholds and document format with the compliance function
- Agree a standard reporting cadence and distribution list
- Evaluate reuse across other business units and customer environments

Note: NetPulse provides the underlying data, calculations and document structure. Regulatory thresholds and report formats remain subject to validation by the compliance function against the currently applicable circulars.